1.1 HTML Elements

HTML elements provide formatting information on how a web page should be displayed. Elements are a request for something to occur. Since HTML documents are stored as ASCII text files they are not compiled but are simply interpreted. If during interpretation the web browser doesn't understand a particular element instruction or is unable to perform it, for whatever reason, then the element is simply ignored. No error message will be displayed.

- It is platform independent. This means that the text and the content are encoded in a way that they can be displayed on a wide range of computers.
- Pages are made of text, images and URL links.
- HTML is structured i.e. it has a beginning, body and end.
- HTML is composed of **tags or elements** which are always enclosed in angle brackets <>
- Tags in HTML are not case sensitive.
- In HTML there are two types of tags, **container or open** tags, and **empty** tags.
- Container tags occur in pairs. An example of a container tag is the <title></title> tag. Whatever is contained within this tag is assigned to the title. Notice that the closing tag has a slash in it.
- Empty tags require no closing tag. An example of an empty tag is the break tag
. This forces the cursor to a new line.
- Tags should always be balanced hence containers should be nested within each other.
- Start tag switches format on
- End tag switches format off
- Start tag <>
- End tag </>
- Example <I> text </I>
- switches italics on and off
- Tag names are not case sensitive
- – but use upper case
- Cannot have spaces after <
- < I> is not <I>
- Can have spaces and other text before >
- Start tags can have attributes
- <tag name attribute name>
- <tag name attribute name="argument">
- Tags can be intermixed with text

Advantages of HTML

1. It can be written in any editor.
2. It is universal and simple to learn and implement.
3. It gives an opportunity to further explore and add more features.

1.2 Software to create Web pages

To create a web page we need a text editor which can save files as ASCII text.
Any editor e.g. MS Word will do but the preferred editor is notepad

1.3 Naming Conventions

There are a number of naming conventions which should be adhered to when creating Web pages. Web pages should be identified by use of the extension .html.
The html language supports a limited number of graphics formats. The most commonly used among these are;

i. Graphics interchange format (.GIF)
ii. JPEG compressed files. (.JPEG or .JPG) extension.

Various other file formats have standard filename extensions, some of the common ones include:

i. .MPG - Em-peg movie file
ii. .mpeg – Em-peg movie file
iii. .WAV – Wave sound file
iv. .TXT – Text file
v. .EXE – DOS executable file
vi. .ZIP – DOS compressed zipped file
vii. .Z – UNIX compressed file
viii. .EPS – Encapsulated Postscript file

1.4 Introduction to the heading Section

The heading section of a HTML document contains general information about the document. Such information includes the title of the document, its relationship to other documents, whether it is a searchable index or not. Information in the Heading section is not displayed in the main hypertext window. Examples of elements in the heading section;

1.4.1 <TITLE> - This element defines the title of the HTML document. This is displayed by many browsers, on the top title bar of the application window.

1.4.2 <ISINDEX> - This element indicates whether the Web document is searchable. It's not used often.

1.4.3 <NEXTID> - This element is used to supply certain web page creation programs with a unique identification number when editing a suite of Web documents. . It's not used often.

1.4.4 <LINK> - This element allows a relationship to be defined between documents which are similar in a Web system.

1.4.5 <BASE> - This element allows the Base URL of the document to be recorded. The base URL is the original location of the document system.

1.4.6 Comments can be inserted into a HTML document anywhere but it is recommended that they be inserted in the heading section of the document and define the name of the document, who created it and when it was written, as a minimum amount of information. Comments are not displayed by the web browser and are included for information only.

Example on how to use comments

```
<HTML>
<HEAD>
<!- -this is my first page- ->
<!--html is easy to use and understand-->
</HEAD>
<BODY>
```

The text of the page goes here
</BODY>
</HTML>

1.5 Structure of a HTML Document

```
<HTML>
<HEAD>
<TITLE> title text goes here </TITLE>
</HEAD>
<BODY>
The text of the page goes here
</BODY>
</HTML>
```

1.6 Exercise

1. Illustrate the use of the <Title> tag in HTML document.

Design an HTML document with the elements of the heading section discussed above. Note that all the elements are open or container tags (elements) and must have their corresponding closing tags